

Physical AI Understanding & Building the Physical World

The **1st Workshop on Physical AI** brings together researchers across vision, graphics, robotics, and generative modeling to advance physics-grounded understanding of the real world — from physical property estimation and 3D/4D reconstruction to differentiable simulation and physically plausible generation. **10 invited talks · 4D reconstruction challenge.**

VENUE

ECCV 2026 · Malmö, Sweden

DATE

September 9, 2026

FORMAT

10 Talks · 4D Challenge

TOPICS OF INTEREST

- Physical scene understanding
- Physics-aware 3D/4D reconstruction
- Generative world models & world simulators
- Spatial & physical reasoning
- Embodied AI & robot learning
- Benchmarks & datasets

COMPETITION

PhysAI Dynamic 4D Reconstruction Challenge

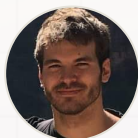
A new synthetic 4D benchmark. **USD 2,000 winner award, supported by Ropedia** + joint technical report.

TRACK 1 4D Reconstruction & Tracking

TRACK 2 Dynamic Video Novel View Synthesis

INVITED SPEAKERS (TENTATIVE · ALPHABETICAL)

Andreas Geiger
Tübingen



David Novotny
Spatial AI



Haiwen Feng
UC Berkeley



Jiajun Wu
Stanford



Kristen Grauman
UT Austin



Marc Pollefeys
ETH Zurich



Matthias Nießner
TU Munich



Mike Roberts
Adobe Research



Ming-Yu Liu
NVIDIA



Vincent Sitzmann
MIT

ORGANIZED BY

Oxford VGG · NTU PVG · NTU MMLab · ETH Zurich · Google DeepMind · NAVER LABS Europe · Ropedia

Join the challenge & the conversation.

Contact: yushi@robots.ox.ac.uk

Full details, schedule & challenge → [workshop website](#)

PhysAI 2026

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